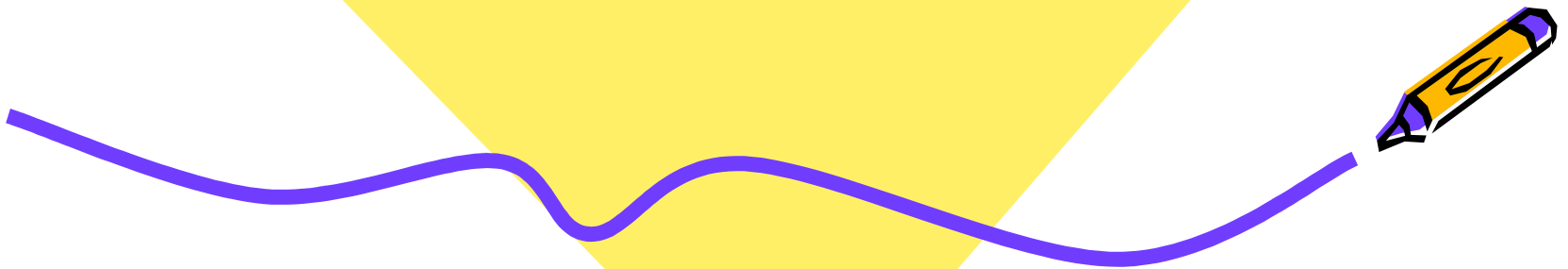




Diabetes mellitus (DM)



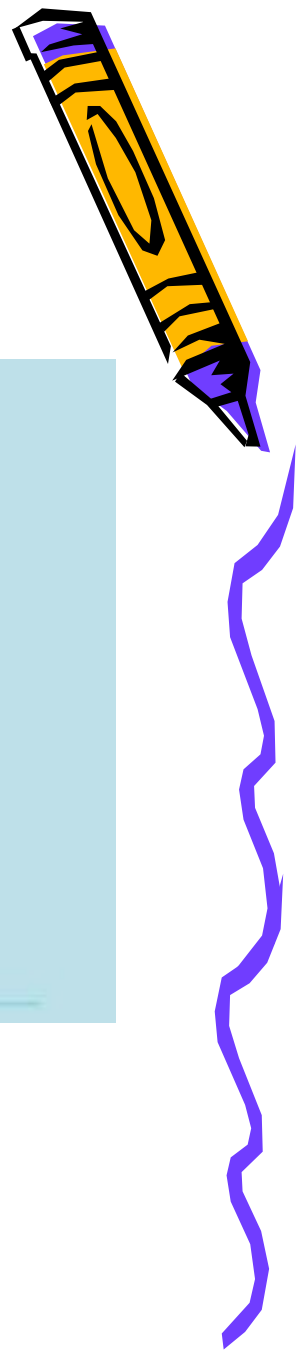
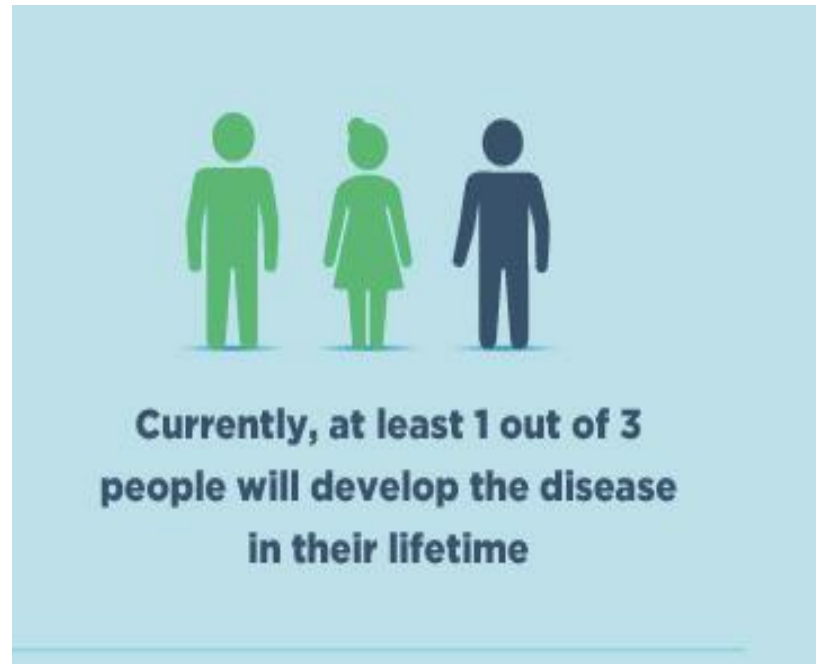
Definition

- Diabetes mellitus (DM) is chronic metabolic disorder caused by absolute or relative deficiency of insulin
- The Insulin is produced by β cells of islets of Langerhans located in the pancreas, and the absence, destruction of these cells results in type 1 diabetes (insulin-dependent diabetes mellitus IDDM)
- Most children with diabetes have IDDM and a lifetime dependence on exogenous insulin
- Most patients with type 2 DM (NIDDM) have insulin resistance



Types of diabetes:

- Type 1
- Type 2
- Gestational diabetes
- Pre diabetes



Type 1 diabetes:

- Also known as juvenile diabetes
- Usually diagnosed in children and young adults
- When body's own immune system destroys the insulin producing cells of the pancreas – beta cells – which produce insulin
- Only 5% of people have this disease
- Body does not produce insulin
- Is not preventable
 - No primary intervention
- Causes?
 - Predisposition to diabetes – genetics - and something (i.e. weather, virus ... etc) in environment triggers the disease



In adults, type 1
diabetes accounts
for approximately

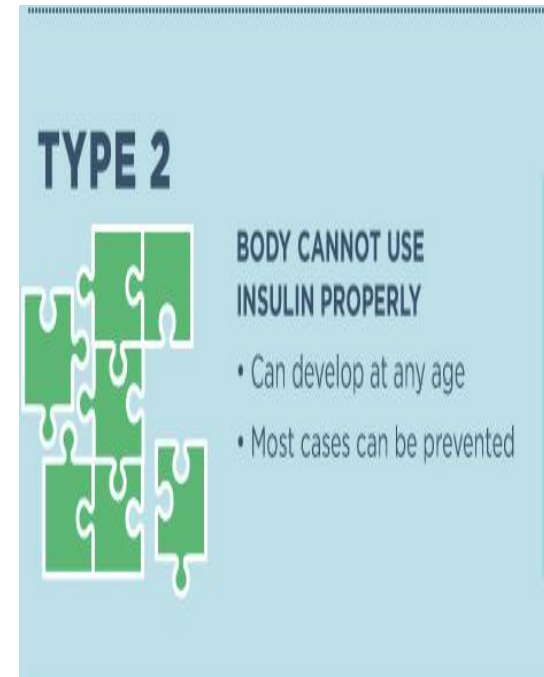
5%

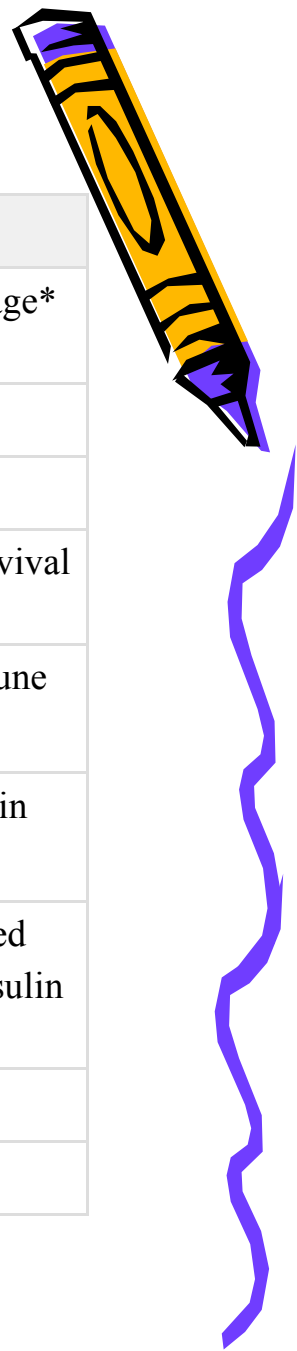
of all diagnosed
cases of diabetes



Type 2 diabetes:

- Most common form of diabetes – about 90% of cases
- Used to be called adult onset, non insulin dependent diabetes
- Body produces insulin, but does not use it properly
 - glucose doesn't enter cells but remains in the bloodstream





	Type 1 diabetes	Type 2 diabetes
Phenotype	Onset primarily in childhood and adolescence	Onset predominantly after 40 years of age*
	Often thin or normal weight	Often obese
	Prone to ketoacidosis	No ketoacidosis
	Insulin administration required for survival	Insulin administration not required for survival
	Pancreas is damaged by an autoimmune attack	Pancreas is not damaged by an autoimmune attack
	Absolute insulin deficiency	Relative insulin deficiency and/or insulin resistance
	Treatment: insulin injections	Treatment: (1) healthy diet and increased exercise; (2) hypoglycemic tablets; (3) insulin injections
Genotype	Increased prevalence in relatives	Increased prevalence in relatives
	HLA association: Yes	HLA association: No



Causes



- 1-Viral infections e.g., mumps, rubella
- 2-Nutritional factors e.g. cow's milk in infancy
- 3-Recent evidence suggests role for vitamin D deficiency in pathogenesis
- 4- ? Toxic chemicals e.g. food preservatives
- 5-Currently, autoimmunity is considered the major factor in pathogenesis of type 1 DM
- 6- Approximately 85% of patients have circulating islet cell antibodies (ICA) & insulin autoantibodies (IAA)



Other causes:

- 1- Congenital absence of the pancreas or islet cells
- 2- Post pancreatectomy
- 3- IDDM secondary to pancreatic damage (i.e., cystic fibrosis, chronic pancreatitis, Thalassemia major)
- 4- Chromosomal disorders such as Down syndrome, Turner syndrome, Klinefelter syndrome



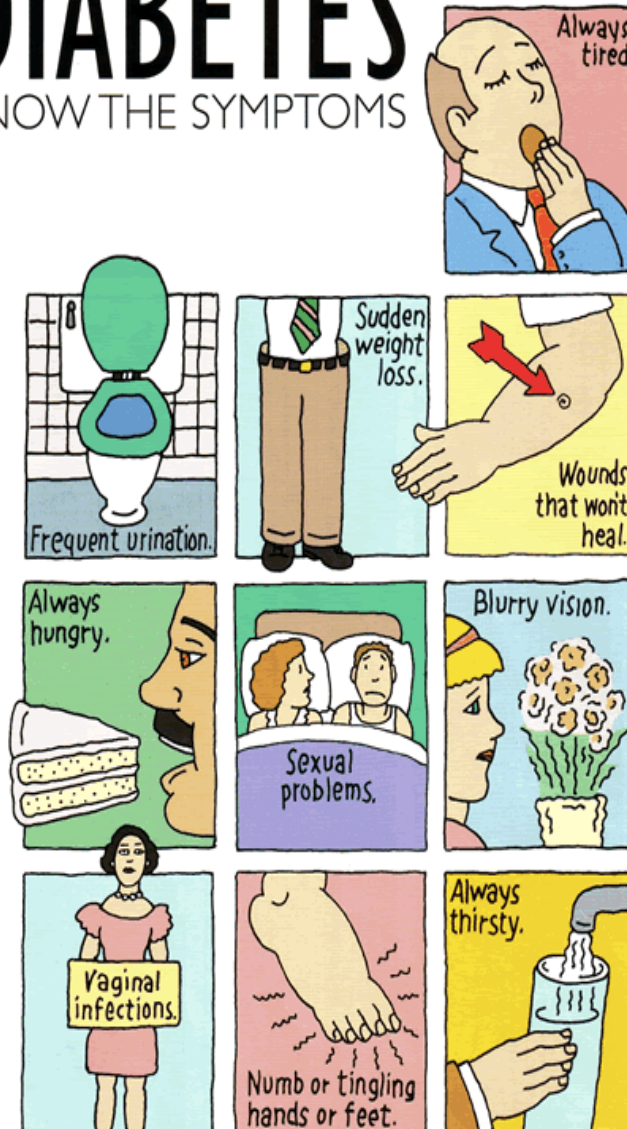
Symptoms of Diabetes:

Common symptoms of diabetes include:

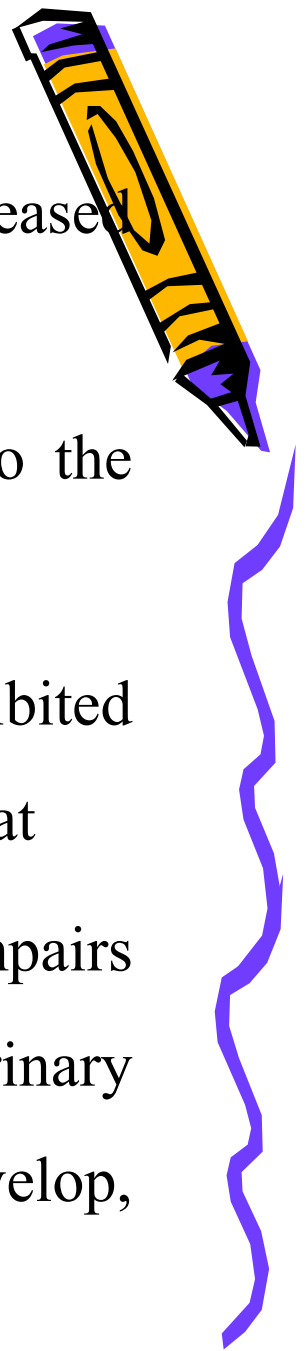
- Excessive thirst and appetite
- Increased urination (sometimes as often as every hour)
- Unusual weight loss or gain
- Fatigue
- Nausea, perhaps vomiting
- Blurred vision
- In women, frequent vaginal infections
- In men and women, yeast infections
- Dry mouth
- Slow-healing sores or cuts
- Itching skin, especially in the groin or vaginal area

DIABETES

KNOW THE SYMPTOMS



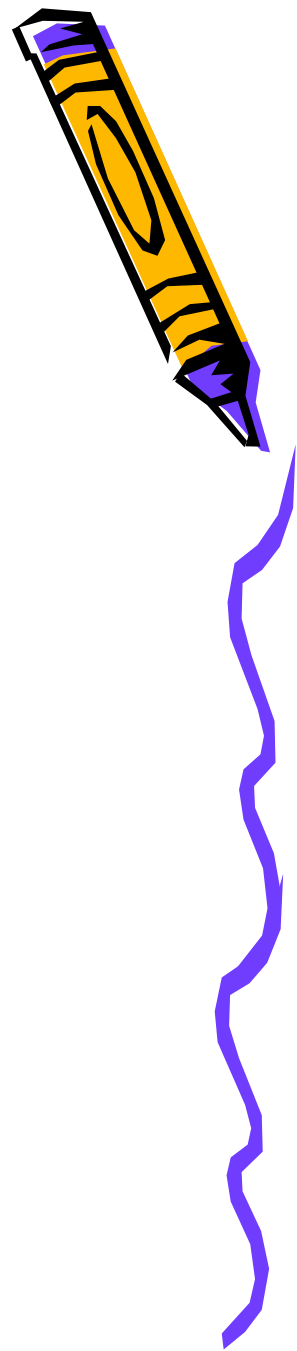
The main symptoms of hyperglycemia



- 1- osmotic diuresis & Glucosuria which leads to increased urinary frequency & volume (polyurea)
- 2- Polydypsea: Increased thirst, which is secondary to the osmotic diuresis causing dehydration
- 3- Weight loss: Insulin deficiency leads to uninhibited gluconeogenesis, causing breakdown of protein and fat
- 4- Other nonspecific symptoms: Hyperglycemia impairs immunity & recurrent infection, particularly of the urinary tract, skin, and respiratory tract. Candidiasis may develop, especially in mouth, groin.



Criteria for the Diagnosis of Diabetes



- ☐ Symptoms of diabetes plus random plasma glucose concentration >200 mg/dl (11.1 mmol/l)
 - The classic symptoms of diabetes include:
polyuria, polydipsia, and unexplained weight loss
- ☐ Fasting glucose >126 mg/dl (7.0 mmol/l)
- ☐ 2-h PPG > 200 mg/dl (11.1 mmol/l) during OGTT
 - The test should be performed by using a glucose load containing equivalent of 75-g anhydrous glucose dissolved in water
- ☐ Hemoglobin A1c $> 6.5\%$



Biochemical changes in diabetes:



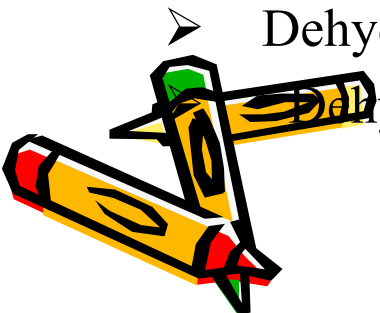
- ❑ Insulin deficiency causes disturbance in carbohydrates, proteins and lipid metabolism.

1- **carbohydrates metabolism:** Insulin deficiency leads to decreased glucose uptake by tissues, decreased glucose oxidation, increased glycogenolysis and increased gluconeogenesis. This leads to:



- a- hunger pain & polyphagia
- b- glucosuria and polyurea
- c- hyperglycemia and increased blood osmolarity leading to dehydration:

- Dehydration of brain cells leads to hyperglycemic coma
- Dehydration of body cells leads to thirst sensation and polydipsia



2- **proteins metabolism:** Insulin deficiency leads to increased protein breakdown leading to muscle wasting, hyperphosphatemia and decreased antibody synthesis (impaired immunity).

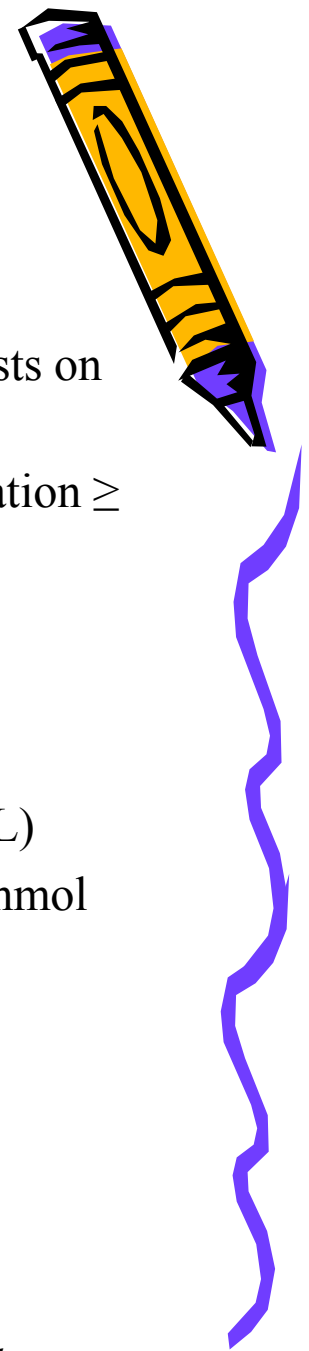
3- **lipid metabolism:** Insulin deficiency leads to excessive lipolysis in adipose tissue and mobilization of free fatty acids and glycerol to blood leading to:

- Weight loss
- Hyperlipidemia , atherosclerosis and fatty liver
- Excess ketone bodies formation, ketonemia causing ketotic coma and hyperkalemia.

4- **Diabetic microangiopathy:** (nephropathy, retinopathy , neurapathy)



Criteria for the Diagnosis of Diabetes Mellitus and Impaired Glucose Homeostasis



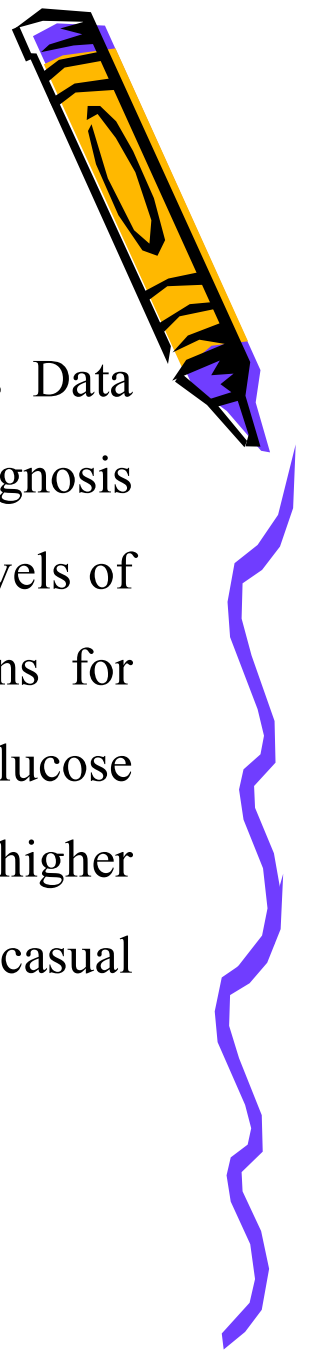
- ❑ **Diabetes mellitus**—positive findings from any two of the following tests on different days:
 - Symptoms of diabetes mellitus* plus random plasma glucose concentration ≥ 200 mg per dL (11.1 mmol per L) *or*
 - FPG ≥ 126 mg per dL (7.0 mmol per L) *or*
 - 2hrPPG ≥ 200 mg per dL (11.1 mmol per L) after a 75-g glucose load
- ❑ **Impaired glucose homeostasis:**
 - Impaired fasting glucose: FPG from 110 to <126 (6.1 to 7.0 mmol per L)
 - Impaired glucose tolerance: 2hrPPG from 140 to <200 (7.75 to <11.1 mmol per L)
- ❑ **Normal**
 - FPG <110 mg per dL (6.1 mmol per L)
 - 2hrPPG <140 mg per dL (7.75 mmol per L)



FPG = fasting plasma glucose;

2hrPPG = two-hour postprandial glucose.

Symptoms include polyuria, polydipsia or unexplained weight loss.



The oral glucose tolerance test (OGTT)

The OGTT previously recommended by the National Diabetes Data Group has been replaced with the recommendation that the diagnosis of diabetes mellitus be based on two fasting plasma glucose levels of 126 mg per dL (7.0 mmol per L) or higher. Other options for diagnosis include two two-hour postprandial plasma glucose (2hrPPG) readings of 200 mg per dL (11.1 mmol per L) or higher after a glucose load of 75 g (recommended by WHO) or two casual glucose readings of 200 mg per dL (11.1 mmol per L) or higher.



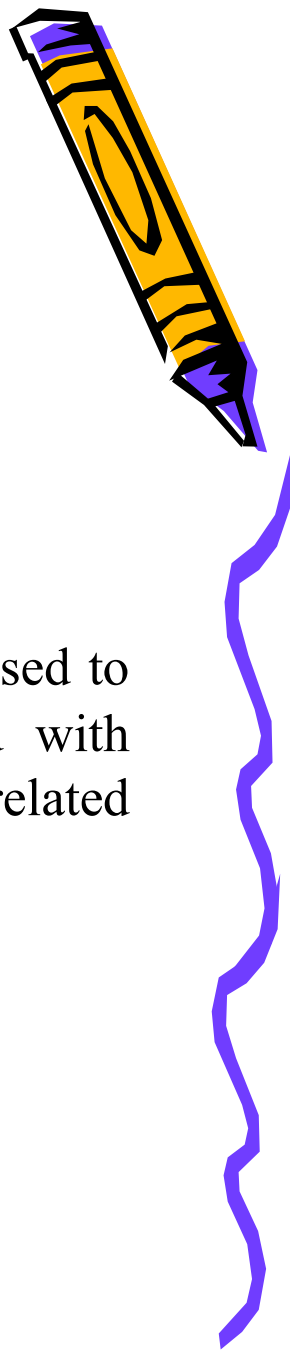
Follow up of diabetes:

- ☐ 2hrPPG &
- ☐ glucosuria &
- ☐ glycated proteins:

1 - Hemoglobin A1c (HbA1c):

Measurements of glycated hemoglobin have commonly been used to monitor the glycemic control of persons already diagnosed with diabetes mellitus. Glycated hemoglobin levels are as highly correlated to adverse clinical outcomes (e.g., retinopathy)

2 - glycated albumin





Complications for uncontrolled diabetes:

Major COMPLICATIONS from diabetes



Wounds in foot that won't heal, leading to **AMPUTATION**

HEART DISEASE

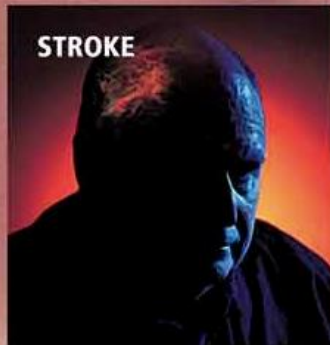


Damaged blood vessels in retina which can cause **BLINDNESS**

KIDNEY FAILURE



STROKE



Chronic complications of diabetes



Diabetic ketoacidosis



- Diabetic ketoacidosis (DKA) is a potentially life-threatening complication of diabetes mellitus. It happens most often in those with type 1 diabetes but can also occur in those with other types of diabetes under certain circumstances.
- Triggers may include infection, not taking insulin correctly, stroke and certain medications such as steroids.
- DKA results from a shortage of insulin; in response, the body switches to burning fatty acids, which produces acidic ketone bodies . DKA is typically diagnosed when testing finds high blood sugar, low blood pH and ketoacids in either the blood or urine.



Symptoms of ketoacidosis

- Severe dehydration
- Smell of Acetone
- Abdominal pain
- Vomiting
- Drowsiness & coma

The primary treatment of DKA:

- intravenous fluids and insulin
- Potassium is usually needed to prevent the development of low blood potassium
- Antibiotics may be required in those with an underlying infection
- Sodium bicarbonate may be given for those with severely low blood

pH

